Semester III **B. Tech CSE (AI)**

BCSAI 302: PROBABILITY AND RANDOM VARIABLES

3L + 0T + 0P + 3C **MM 100**

Unit 1: Probability And Random Variables

Probability – Axioms of probability – Conditional probability – Baye‘s theorem – Discrete and continuous random variables – Moments – Moment generating functions – Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.

Unit 2: Two – Dimensional Random Variables

Joint distributions – Marginal and conditional distributions – Covariance – Correlation and linear regression – Transformation of random variables – Central limit theorem (for independent and identically distributed random variables).

Unit 3: Random Processes

Classification – Stationary process – Markov process – Markov chain – Poisson process – Random telegraph process.

Unit 4: Correlation And Spectral Densities

Auto correlation functions – Cross correlation functions – Properties – Power spectral density – Cross spectral density – Properties.

Unit 5: Linear Systems With Random Inputs

Linear time invariant system – System transfer function – Linear systems with random inputs – Auto correlation and cross correlation functions of input and output

Text /Reference Books:

1. Ibe, O.C.,” Fundamentals of Applied Probability and Random Processes “, 1st Indian Reprint, Elsevier, 2007.
2. Peebles, P.Z., “Probability, Random Variables and Random Signal Principles “, Tata McGraw Hill, 4th Edition, New Delhi, 2002.
3. Cooper. G.R., McGillem. C.D., “Probabilistic Methods of Signal and System Analysis”, Oxford University Press, New Delhi, 3rd Indian Edition, 2012.
4. Hwei Hsu, “Schaums Outline of Theory and Problems of Probability, Random Variables and Random Processes “, Tata McGraw Hill Edition, New Delhi, 2004.